CHEN HONGYU

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EDUCATION

The Hong Kong University of Science and Technology(HKUST) **Hong Kong**
BSc in Mathematics(General Mathematics) and BSc in Computer Science
Expected Graduation Date: Jun 2027
CGA 3.952/4.3 (Top 3%)
Dean's List: 2023 Fall, 2024 Spring,2024 Fall,2025 Spring,2026 Spring
Relevant Courses: Linear Algebra(A+), Probability(A+), Discrete Mathematical Tools for Computer Science(A+), Introduction to Computer Science(Python,A+,C++, A), Calculus(A+),Multi-variable Calculus(A), Introduction to AI(A), Large-Scale Machine Learning System for Foundation Models(A), Operating System(A).

École Polytechnique Fédérale de Lausanne(EPFL) **Switzerland**
Exchange student in Mathematics ***Sep 2025-Jan 2026***
Core course: Machine Learning(5.5/6), Algebra(5.5/6)

Guangzhou Second Middle School **Guangzhou**
Top 0.25% in Guangdong Gaokao *Graduation Date: Jun 2023*

EXTRA-CURRICULAR ACTIVITIES

School of Science, HKUST **Hong Kong**
Peer Mentor in SSCI MAGET Peer Mentor Program **Sep 2024 - Jan 2025**
• Successfully enhanced communication, leadership, and empathy abilities through regular interactions with mentees, fostering a supportive and collaborative environment
• Built a diverse and supportive network of professional contacts and friendships by connecting with mentees and fellow mentors, which has been instrumental for both personal and career growth

Student Ambassador,HKUST **Hong Kong**
Dec 2024 - Jan 2024 **Sep 2025 - Present**
• Enhanced public speaking and communication abilities by presenting university information to prospective students, demonstrating confidence and the ability to convey complex information clearly.
• Demonstrated leadership and the ability to positively influence others by actively participating in information sessions, providing valuable insights and guidance to future students.

PROJECT EXPERIENCE

LLM-Assisted Multimodal Sensing on Mobile Devices,HKUST **Hong Kong**
Research Assistant **Jun 2025 - Dec 2025**
· Full-Stack Development: Developed a mobile App by Android studio and Python/SQL backend to capture and process high-frequency IMU and system data with sub-millisecond synchronization.
· Data Classification: Engineered automated pipelines to clean and classify massive sensor datasets, ensuring high data integrity through advanced signal processing.
· Data Synthesis & Augmentation: Spearheaded a data synthesis engine to merge heterogeneous streams and generate synthetic samples, prepare a more large dataset for LLM to predict people's action.

EPFL ML Road Segmentation Competition **Switzerland , Nov 2025 - Dec 2025**
· Implemented and optimized U-Net and SegNet architectures for pixel-level road segmentation, achieving high-precision feature extraction from complex imagery.
· Engineered a robust pipeline for synthetic data generation and image processing, significantly expanding the training dataset to enhance model generalization.

Zero-Shot Visual-language feature alignment via ViT-based Contrastive Learning Problem ,HKUST **Hong Kong**
Course Project Leader **Feb 2026-May 2026**
· Developed a Multi-modal CLIP-based Framework with ViT Backbone: Engineered a dual-tower architecture from scratch, aligning Vision Transformer image embeddings with text representations in a shared latent space using contrastive learning.
· Implemented Zero-Shot Classification and Image Retrieval Systems: Leveraged symmetric cross-entropy loss to enable "open-vocabulary" image categorization and high-dimensional vector search without the need for task-specific fine-tuning.

Simulation Data Pipeline and Architecture Adaptation for Navigation World Models, HKUST **Hong Kong**
Research Intern **April 2026–Present**

· **Developed a Perturbation-Based Simulation Data Pipeline:** Built a pipeline in NVIDIA Isaac Sim integrating NavDP and A* planning to collect navigation trajectories and visual observations, with configurable perturbations to explore collisions, getting stuck, and recovery behaviors for world-model research.

· **Adapted the World-Model Architecture for Navigation:** Modified cosmos3 components to better support navigation-oriented visual dynamics modeling using simulation-generated trajectories.

LANGUAGES, SKILLS & INTERESTS

Skills: Microsoft Office (Excel, Word, PowerPoint), CHATGPT/Prompt, Programming(Python, C++) and Data Processing, Pytorch, Latex, Computer Vision, Machine Learning, Reinforcement Learning.
Languages: Chinese(Native), Cantonese(Fluent), English(Fluent, IELTS 7.0)